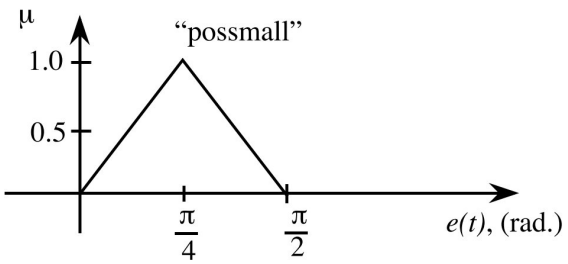
Exercises 2026-09-17

(i) Write three-five fuzzy rules that determine heart attack risk, using: Three 'universes of discourse' (UoD): diet, exercise, and risk. Use two or three fuzzy sets per UoD, and their membership functions (represent graphically).

(ii) The "support" of a fuzzy set with membership function $\mu(x)$ is the (crisp) set of all points x on the universe of discourse such that $\mu(x) > 0$ and the "α-cut" is the (crisp) set of all points on the universe of discourse such that $\mu(x) > \alpha$. What is the support and 0.5-cut for the fuzzy set shown below?

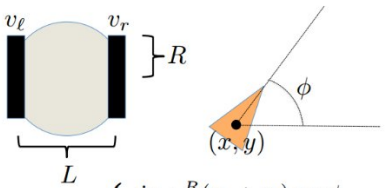


(iii) The "height" of a fuzzy set with membership function $\mu(x)$ is the highest value that $\mu(x)$ reaches on the universe of discourse on which it is defined. A fuzzy set is said to be "normal" if its height is equal to one. What is the height of the fuzzy set shown in the figure above?

(iv) Give an example of a fuzzy set that is normal and one that is not normal.

(v) A linguistic "hedge" is a modifier to a linguistic value such as "*very*" or "*more or less*." When we use linguistic hedges for linguistic values that already have membership functions, we can simply modify these membership functions so that they represent the modified linguistic values. Consider the membership function in the figure above. Suppose that we obtain the membership function for "error is very possmall" from the one for "possmall" by squaring the membership values (i.e., $\mu_{verypossmall} = (\mu_{possmall})^2$). Sketch the membership function for "error is very possmall." For "error is more or less possmall", we could use $\mu_{moreorlesspossmall} = \sqrt{\mu_{possmall}}$. Sketch the membership function for "error is more or less possmall."

(vi) Assume that you have a differential drive robot (L = 20 cm) and that you can control the speeds of the left and right wheel, v_r and v_l (+/- 20 cm/s):



$$\begin{cases} \dot{x} = \frac{R}{2}(v_r + v_\ell)\cos\phi \\ \dot{y} = \frac{R}{2}(v_r + v_\ell)\sin\phi \\ \dot{\phi} = \frac{R}{L}(v_r - v_\ell) \end{cases}$$

Assume that the robot always knows its location (x, y) and orientation (φ) in the global frame of reference. The robot has five non-overlapping obstacle sensors that cover [-π/2, π/2] with respect to current heading of the robot. Each sensor has a range of 2m and returns the distance of the closest obstacle detected in that range, or a reading of 2m in case

no obstacle is detected. Try to design a fuzzy logic-based controller, that enables the robot to move from its initial location to a given goal location (x_g, y_g) while avoiding obstacles. Hence, you should define a rule-base, linguistic variables, linguistic values, and membership functions. It is suggested that you base your initial controller on a simple Bug algorithm without memory. Afterwards, try to extend your controller and consider how Bug algorithms that rely on memory could be implemented, while still using fuzzy control – at least partially.